

TUAPSE, Russian Federation

WMO: 370180

Lat: 44.100N

Lon: 39.067E

Elev: 202

StdP: 14.59

Time Zone: 3.00 (E03)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	27.1	7.5	8.1	25.4	13.3	10.9	29.4	25.8	48.9	23.8	46.8	12.1	40	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	90.3	74.9	87.7	74.3	85.4	73.2	78.3	86.3	76.7	84.3	75.2	83.1	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.9	136.5	84.5	74.2	128.8	82.9	72.6	121.5	81.3	42.1	85.9	40.4	84.5	38.9	83.1	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	19.8	16.7	18.3	94.3	5.3	3.3	14.5	96.7	11.4	98.6	8.4	100.5	4.5	102.9
			15.4	80.2	5.0	2.3	11.8	81.9	8.8	83.3	6.0	84.6	2.4	86.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	42.4	43.3	47.3	54.7	63.5	70.4	76.3	77.6	70.4	60.6	52.0	44.9	(6)
	DBStd	13.93	7.50	7.46	6.33	6.24	6.16	4.83	4.54	4.23	5.49	6.37	7.14	8.01	(7)
	HDD50	864	246	204	129	23	0	0	0	0	0	6	62	194	(8)
	HDD65	3479	700	608	549	315	102	11	0	0	17	162	391	624	(9)
	CDD50	4042	10	17	45	163	418	613	815	856	612	336	121	36	(10)
	CDD65	1181	0	0	0	5	56	174	350	391	179	26	0	0	(11)
	CDH74	8590	0	0	0	19	286	859	2904	3480	965	77	0	0	(12)
	CDH80	2456	0	0	0	1	58	151	881	1156	204	5	0	0	(13)
(14) Wind	WSAvg	6.9	8.6	8.4	7.8	6.5	5.1	5.3	5.3	5.6	6.4	7.0	8.0	8.7	(14)
(15) Precipitation	PrecAvg	60.2	6.8	5.1	4.7	3.7	4.1	4.2	3.8	3.7	4.9	6.1	6.2	7.0	(15)
	PrecMax	77.8	10.4	10.6	10.6	7.3	10.9	9.7	13.2	10.6	14.9	15.3	13.0	14.3	(16)
	PrecMin	41.0	2.0	1.5	2.2	1.4	0.6	0.9	0.2	0.2	0.7	1.1	0.8	1.3	(17)
	PrecStd	8.2	2.2	2.4	1.9	1.6	2.6	2.0	3.1	3.0	3.0	3.3	3.3	2.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	61.0	67.7	76.8	85.9	87.5	93.4	94.4	89.4	80.1	70.4	64.8	(19)
		MCWB	47.8	47.8	52.2	59.0	66.8	71.2	76.8	76.7	74.1	66.8	56.3	53.9	(20)
	2%	DB	55.9	58.0	63.1	71.6	80.4	83.9	90.0	90.9	85.1	76.0	67.2	61.0	(21)
		MCWB	47.2	47.4	49.8	56.5	66.1	71.7	75.1	75.5	71.5	64.5	55.8	50.9	(22)
	5%	DB	54.2	55.8	59.4	67.7	76.8	81.2	87.4	88.2	82.0	73.1	64.7	58.3	(23)
		MCWB	46.6	46.9	48.9	55.5	64.8	71.1	74.6	74.8	70.0	63.0	55.5	49.5	(24)
	10%	DB	52.1	53.2	56.4	64.5	73.5	78.7	84.9	85.9	79.4	70.2	62.2	55.7	(25)
		MCWB	45.9	45.9	48.0	54.2	63.9	70.0	73.6	73.8	69.0	62.4	54.7	48.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	50.6	53.7	61.1	70.3	76.0	80.6	80.6	77.7	68.4	60.7	56.1	(27)
		MCDB	55.0	55.5	62.7	70.4	78.8	82.8	89.0	90.8	84.6	76.2	65.2	63.1	(28)
	2%	WB	49.2	49.3	51.7	58.8	68.1	73.9	78.4	78.6	73.7	66.5	58.9	52.7	(29)
		MCDB	53.4	54.7	59.7	67.8	76.0	81.2	86.0	87.0	82.3	73.6	64.1	57.9	(30)
	5%	WB	48.0	48.1	50.2	56.9	66.4	72.4	76.8	77.0	71.6	64.6	57.2	51.0	(31)
		MCDB	52.4	53.6	56.6	65.3	74.5	79.5	84.5	84.7	79.9	70.9	62.5	56.3	(32)
	10%	WB	46.6	46.8	48.9	55.4	64.8	70.9	75.3	75.5	69.9	63.0	55.4	49.4	(33)
		MCDB	51.4	52.3	54.7	62.7	72.2	77.5	82.9	83.6	77.8	68.9	61.0	54.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.8	10.1	11.8	12.8	12.2	13.3	14.4	14.2	13.0	12.1	10.0	(35)
		MCDBR	11.2	12.4	15.4	17.7	18.3	15.1	16.1	17.2	17.6	17.0	16.0	12.2	(36)
	5% WB	MCWBR	7.3	7.1	7.8	8.4	8.8	8.6	9.3	9.9	10.4	10.0	10.6	8.4	(37)
		MCWBR	9.3	9.7	12.5	14.4	15.3	13.3	14.1	14.8	16.0	14.6	13.6	10.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.350	0.378	0.424	0.462	0.430	0.429	0.446	0.447	0.407	0.389	0.358	0.345	(40)
		taud	2.318	2.236	2.137	2.081	2.271	2.324	2.290	2.283	2.370	2.369	2.385	2.352	(41)
	Ebn at Noon		238	252	254	255	268	268	262	257	259	246	235	228	(42)
		Edh at Noon	26	34	43	49	42	40	41	40	34	30	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		417	684	984	1483	1846	2059	2099	1854	1389	902	551	371	(44)
	RadStd		57	83	84	164	202	136	106	115	136	83	64	49	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+126

Nomenclature: See separate page